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PAGE 1: INTRODUCTION

1. Introduction to Inequalities

Inequalities are fundamental mathematical statements that compare the relative sizes of two expressions using symbols such as $<$ (less than), $>$ (greater than), $\leq$ (less than or equal to), and $\geq$ (greater than or equal to). Unlike equations, which assert equality between two expressions, inequalities establish bounds and relationships that are crucial in many areas of mathematics.

1.1 Definition and Basic Concepts

An inequality is a mathematical relation that expresses the order between two values or expressions. For real numbers a and b :

- $a < b$ means "a is strictly less than b"
- $a > b$ means "a is strictly greater than b"
- $a \leq b$ means "a is less than or equal to b"
- $a \geq b$ means "a is greater than or equal to b"

1.2 Importance of Classical Inequalities

Classical inequalities are a set of powerful, generalized inequalities that have wide applications in:

Mathematical Analysis: Proving convergence, boundedness, and continuity of functions. They provide essential tools for establishing limits and studying the behavior of sequences and series.

Optimization Theory: Finding minimum and maximum values of functions without using calculus. Inequalities often provide simpler and more elegant solutions than derivative-based methods.

Mathematical Competitions: Essential tools for Olympiad problems, including the International Mathematical Olympiad (IMO) and Putnam Competition.

Probability and Statistics: Establishing bounds on probabilities, variance, and expected values. Many statistical results rely fundamentally on classical inequalities.

Physics and Engineering: Error analysis, approximation bounds, and stability analysis in physical systems.

Number Theory: Proving results about integers, prime numbers, and Diophantine equations.

1.3 Types of Inequalities

Inequalities can be classified into several categories:

1. **Algebraic Inequalities:** Involve algebraic expressions and operations
2. **Geometric Inequalities:** Relate to geometric quantities like lengths, areas, and volumes
3. **Integral Inequalities:** Involve integrals and are fundamental in analysis
4. **Differential Inequalities:** Involve derivatives and differential equations
5. **Combinatorial Inequalities:** Relate to counting and discrete structures

This assignment focuses primarily on algebraic classical inequalities, which form the foundation for all other types.

1.4 Key Properties of Inequalities

Understanding the basic properties of inequalities is essential:

Transitivity: If $a \leq b$ and $b \leq c$, then $a \leq c$

Additivity: If $a \leq b$, then $a + c \leq b + c$ for any c

Multiplicativity: If $a \leq b$ and $c > 0$, then $ac \leq bc$; if $c < 0$, then $ac \geq bc$

Reciprocal Property: If $0 < a \leq b$, then $1/a \geq 1/b > 0$

Square Property: If $a \geq 0$, then $a^2 \geq 0$, and $a^2 = 0$ if and only if $a = 0$

Triangle Inequality: $|a + b| \leq |a| + |b|$ for all real numbers a, b

1.5 Equality Conditions

One of the most important aspects of classical inequalities is understanding when equality holds. Equality conditions are crucial because:

- They reveal the optimal or extreme cases
- They help verify if a bound is tight
- They often provide insight into the structure of the problem
- In optimization, equality conditions identify optimal solutions

For most classical inequalities, equality holds if and only if all variables are equal, or in specific proportional relationships.

1.6 Proof Techniques

Several fundamental techniques are used to prove classical inequalities:

1. **Mathematical Induction:** Proving for $n = 2$, then extending to arbitrary n
2. **Cauchy Induction:** A specialized form of induction for inequalities
3. **Convexity Arguments:** Using properties of convex functions (Jensen's inequality)
4. **Rearrangement:** Using the rearrangement inequality
5. **Algebraic Manipulation:** Expanding, factoring, and completing squares
6. **Substitution:** Introducing new variables to simplify expressions
7. **Calculus Methods:** Using derivatives to find extrema

This assignment will explore these techniques in detail through specific inequalities.

PAGE 2: HISTORICAL BACKGROUND

2. Historical Background of Inequalities

The study of inequalities has a rich history spanning over 2,300 years, evolving from ancient geometric insights to modern analytical tools.

2.1 Ancient Greek Mathematics (300 BCE)

The formal study of inequalities began with the ancient Greeks, particularly Euclid around 300 BCE. Greek mathematicians developed geometric inequalities as part of their work on geometry and number theory.

Euclid's Contributions: In his seminal work "Elements," Euclid included several geometric inequalities, including the triangle inequality for lengths. The geometric approach to inequalities dominated for centuries.

Archimedes (287-212 BCE): Used inequality methods to bound the value of π , establishing that $3 + 10/71 < \pi < 3 + 1/7$, demonstrating early use of inequalities for approximation.

Aristotle (384-322 BCE): Developed philosophical foundations for understanding order and comparison, which influenced mathematical thinking about inequalities.

2.2 Islamic Golden Age (8th-14th Century)

During the Islamic Golden Age, mathematicians built upon Greek foundations and began developing more explicit algebraic forms of inequalities.

Al-Khwarizmi (780-850): Developed algebraic methods that would later be essential for inequality manipulation.

Omar Khayyam (1048-1131): Contributed to understanding relationships between algebraic expressions, laying groundwork for algebraic inequalities.

2.3 European Renaissance (15th-17th Century)

The Renaissance saw the development of modern algebraic notation, making inequalities more accessible and manipulable.

Robert Recorde (1510-1558): Introduced the equals sign (=) in 1557, which helped distinguish equations from inequalities.

Thomas Harriot (1560-1621): Introduced the modern inequality symbols $<$ and $>$ in the early 17th century, revolutionizing how inequalities were written and understood.

Pierre Bouguer (1698-1758): Contributed to the formalization of inequality notation that we still use today.

2.4 Modern Era (18th-19th Century)

The 18th and 19th centuries saw the development of analysis and the formalization of classical inequalities.

Augustin-Louis Cauchy (1789-1857): Developed the Cauchy-Schwarz inequality (though Schwarz contributed independently), which became one of the most important inequalities in mathematics.

Hermann Schwarz (1843-1921): Independently discovered and popularized the Cauchy-Schwarz inequality in its modern form.

Johan Jensen (1859-1925): Formulated Jensen's inequality in 1906, providing a powerful tool for convex functions.

Otto Hölder (1859-1937): Developed Hölder's inequality in 1889, generalizing the Cauchy-Schwarz inequality.

Hermann Minkowski (1864-1909): Developed Minkowski's inequality in 1896, establishing the triangle inequality for L^p spaces.

2.5 20th Century Developments

G.H. Hardy, J.E. Littlewood, and G. Pólya: Their 1934 book "Inequalities" became the definitive reference on the subject, systematizing the field and introducing many new techniques.

Pólya's Dream: According to mathematical folklore, Pólya discovered an elegant proof of the AM-GM inequality using the convexity of the exponential function in a dream, which he later reported as "the best mathematics he had ever dreamt".

Nesbitt's Inequality (1903): Arthur Nesbitt proposed this inequality in the "Educational Times," and it became a classic problem in mathematical competitions.

Schur's Inequality (1903): Issai Schur developed this inequality while working on number theory, and it became fundamental in olympiad mathematics.

2.6 Contemporary Significance

The quote "Every mathematician loves an inequality" is part of the folklore of mathematics, reflecting the central importance of inequalities in modern mathematical research.

Research Areas: Modern research continues to develop new inequalities and refine classical ones, with applications in:

- Optimization and operations research
- Machine learning and data science
- Quantum mechanics and information theory
- Economics and game theory
- Computer science and algorithm analysis

2.7 Notable Mathematical Quotes

- **G.H. Hardy:** "Inequalities are the essence of analysis"
- **George Pólya:** "An idea which can be used only once is a trick. If one can use it more than once, it becomes a method" (referring to inequality techniques)
- **David Hilbert:** Emphasized the importance of inequalities in his work on functional analysis

2.8 Evolution of Proof Techniques

The methods for proving inequalities have evolved significantly:

Early Methods: Geometric proofs dominated, using visual arguments about areas and lengths.

Algebraic Era: Development of algebraic manipulation techniques in the 17th-18th centuries.

Analytic Methods: Use of calculus and analysis in the 19th century.

Modern Approaches: Combining multiple techniques, computer-assisted proofs, and abstract generalizations.

PAGE 3: AM-GM INEQUALITY

3. Arithmetic Mean-Geometric Mean (AM-GM) Inequality

The AM-GM inequality is the most fundamental and widely used classical inequality. It states that the arithmetic mean of non-negative numbers is always greater than or equal to their geometric mean.

3.1 Statement of the Inequality

For any non-negative real numbers $a_1, a_2, \dots, a_n$:

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}$$

with equality if and only if $a_1 = a_2 = \dots = a_n$.

3.2 Two-Variable Case

For two non-negative real numbers a and b :

$$\frac{a+b}{2} \geq \sqrt{ab}$$

This can be easily proven by observing:

$$(a-b)^2 \geq 0 \implies (a-b)^2 \geq 0$$

$$a^2 - 2ab + b^2 \geq 0 \implies a^2 - 2ab + b^2 \geq 0$$

$$a + b \geq 2ab \implies a + b \geq 2ab$$

$$a + b \geq 2ab \implies a + b \geq 2ab$$

Equality holds if and only if $\sqrt{a} = \sqrt{b}$, i.e., $a = b$.

3.3 Fundamental Application: $a + \frac{1}{a} \geq 2$

Problem: Prove that if $a > 0$, then $a + \frac{1}{a} \geq 2$.

Solution using AM-GM:

Applying AM-GM to the numbers a and $\frac{1}{a}$:

$$a + \frac{1}{a} \geq 2\sqrt{a \cdot \frac{1}{a}} = 2\sqrt{1} = 2$$

$$a + \frac{1}{a} \geq 2 \implies a + \frac{1}{a} \geq 2$$

Equality holds when $a = \frac{1}{a}$, i.e., $a = 1$.

3.4 Proof Methods for AM-GM

Method 1: Mathematical Induction (Cauchy Induction)

Step 1: Prove for $n = 2$ (shown above)

Step 2: Prove for $n = 2^k$ by induction on k

For $n = 4$:

$$\begin{aligned} a_1 + a_2 + a_3 + a_4 &= a_1 + a_2 + a_3 + a_4 \\ &\geq a_1 + a_2 + a_3 + a_4 \\ &\geq a_1 + a_2 + a_3 + a_4 \\ &\geq a_1 + a_2 + a_3 + a_4 \end{aligned}$$

Step 3: For $2^k < n < 2^{k+1}$, pad the sequence with $A = \frac{a_1 + \dots + a_n}{n}$ to reach 2^{k+1} terms.

Method 2: Pólya's Method (Using Convexity of e^x)

This elegant proof, discovered by Pólya in a dream, uses the convexity of the exponential function:

Let $G = \sqrt[n]{a_1 a_2 \dots a_n}$ be the geometric mean. Define $x_i = \ln\left(\frac{a_i}{G}\right)$.

Then $\sum_{i=1}^n x_i = \sum_{i=1}^n \ln\left(\frac{a_i}{G}\right) = \ln\left(\frac{a_1 a_2 \dots a_n}{G^n}\right) = \ln(1) = 0$.

Since e^x is convex, by Jensen's inequality:

$$e^{x_1} + e^{x_2} + \dots + e^{x_n} \geq e^{x_1 + x_2 + \dots + x_n} = e^0 = 1$$

$$a_1 G + a_2 G + \dots + a_n G \geq 1nGa_1 + Ga_2 + \dots + Gan \geq 1$$

$$a_1 + a_2 + \dots + a_n \geq G = a_1 a_2 \dots a_n \Rightarrow a_1 + a_2 + \dots + a_n \geq G = na_1 a_2 \dots a_n$$

Method 3: Using Jensen's Inequality

Since $f(x) = -\ln(x)$ is convex for $x > 0$:

$$-\ln\left(\frac{a_1 + \dots + a_n}{n}\right) \leq -\ln\left(\frac{a_1}{n}\right) + \dots + (-\ln\left(\frac{a_n}{n}\right)) = -\ln\left(\frac{a_1 \dots a_n}{n^n}\right) = -\ln\left(\frac{a_1 \dots a_n}{n^n}\right) = -\ln(a_1 \dots a_n) + n \ln n$$

Taking negative and exponentiating gives AM-GM.

3.5 Weighted AM-GM Inequality

For non-negative numbers $a_1, \dots, a_n$ and non-negative weights $\omega_1, \dots, \omega_n$ with $\sum \omega_i > 0$:

$$\frac{\sum_{i=1}^n \omega_i a_i}{\sum_{i=1}^n \omega_i} \geq \sqrt[n]{\prod_{i=1}^n a_i^{\omega_i}}$$

with equality if and only if all a_i are equal.

3.6 Worked Examples

Example 1: If x, y, z are positive real numbers such that $xyz = 1$, find the minimum value of $4xy^2 + 2x^2y + 27z^3$.

Solution:

Using AM-GM:

$$4xy^2 + 2x^2y + 27z^3 \geq (4xy^2)(2x^2y)(27z^3) = 216x^3y^3z^3 = 6xyz = 6$$

Therefore, $4xy^2 + 2x^2y + 27z^3 \geq 6$.

The minimum value is **6**, achieved when $4xy^2 = 2x^2y = 27z^3$.

Example 2: Prove that for positive real numbers a, b, c :

$$(a+b)(b+c)(c+a) \geq 8abc$$

Solution:

By AM-GM:

- $a + b \geq 2\sqrt{ab}$

- $b + c \geq 2\sqrt{bc}$
- $c + a \geq 2\sqrt{ca}$

Multiplying:

$$(a+b)(b+c)(c+a) \geq 2ab \cdot 2bc \cdot 2ca = 8abc$$

Equality when $a = b = c$.

Example 3: Find the minimum value of $x + \frac{1}{x}$ for $x > 0$.

Solution:

By AM-GM:

$$x + \frac{1}{x} \geq x \cdot \frac{1}{x} = 1$$

$$x + \frac{1}{x} \geq 2$$

Minimum value is **2**, achieved at $x = 1$.

3.7 Applications of AM-GM

1. **Optimization:** Finding minimum/maximum values without calculus
2. **Algebraic Proofs:** Proving other inequalities
3. **Number Theory:** Establishing bounds on number-theoretic functions
4. **Geometry:** Proving geometric inequalities
5. **Mathematical Competitions:** Fundamental tool for Olympiad problems

PAGE 4: CAUCHY-SCHWARZ INEQUALITY

4. Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality is one of the most important and widely applicable inequalities in mathematics, relating a product of sums of squares to the square of a sum of products.

4.1 Statement of the Inequality

For real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$:

$$\left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right) \geq \left(\sum_{i=1}^n a_i b_i\right)^2$$

In vector notation: $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|$.

Equality Condition: Equality holds if and only if $\frac{a_i}{b_i} = k$ for some constant $k \in \mathbb{R}$ for all i where $b_i \neq 0$ (i.e., the sequences are proportional).

4.2 Two-Variable Case

For four real numbers a, b, c, d :

$$(a^2+b^2)(c^2+d^2) \geq (ac+bd)^2$$

Proof:

Expanding both sides:

$$a^2c^2+a^2d^2+b^2c^2+b^2d^2 \geq a^2c^2+2abcd+b^2d^2$$

$$a^2d^2+b^2c^2 \geq 2abcd$$

$$a^2d^2-2abcd+b^2c^2 \geq 0$$

$$(ad-bc)^2 \geq 0$$

This is always true, and equality holds when $ad = bc$, i.e., $\frac{a}{b} = \frac{c}{d}$.

4.3 Proof Methods

Method 1: Quadratic Polynomial Approach

Consider the quadratic polynomial in t :

$$P(t) = \sum_{i=1}^n (a_i t - b_i)^2 \geq 0$$

Expanding:

$$P(t) = \left(\sum_{i=1}^n a_i^2\right)t^2 - 2\left(\sum_{i=1}^n a_i b_i\right)t + \left(\sum_{i=1}^n b_i^2\right) \geq 0$$

Since this quadratic is always non-negative, its discriminant must be non-positive:

$$\Delta = 4\left(\sum_{i=1}^n a_i b_i\right)^2 - 4\left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right) \leq 0$$

Therefore:

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right)$$

Method 2: Linear Algebra Interpretation

In $\mathbb{R}^n$, for vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

Since $|\cos \theta| \leq 1$:

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

This is the Cauchy-Schwarz inequality.

4.4 Titu's Lemma (Cauchy-Schwarz in Engel Form)

A useful special case:

$$\frac{x_1^2}{a_1} + \frac{x_2^2}{a_2} + \cdots + \frac{x_n^2}{a_n} \geq \frac{(x_1 + x_2 + \cdots + x_n)^2}{a_1 + a_2 + \cdots + a_n}$$

for positive $a_1, \dots, a_n$.

This follows from Cauchy-Schwarz by setting $a_i = \sqrt{a_i}$ and $b_i = \frac{x_i}{\sqrt{a_i}}$.

4.5 Worked Examples

Example 1: Given $a^2 + b^2 + c^2 = 1$, find the maximum value of $12a + 4b + 3c$.

Solution:

By Cauchy-Schwarz:

$$\begin{aligned} (12^2 + 4^2 + 3^2)(a^2 + b^2 + c^2) &\geq (12a + 4b + 3c)^2 \\ (144 + 16 + 9)(1) &\geq (12a + 4b + 3c)^2 \\ 169 &\geq (12a + 4b + 3c)^2 \\ -13 &\leq 12a + 4b + 3c \leq 13 \end{aligned}$$

Maximum value is **13**, achieved when $\frac{a}{12} = \frac{b}{4} = \frac{c}{3}$.

Example 2: Prove Nesbitt's Inequality using Cauchy-Schwarz:

For positive real numbers a, b, c :

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

Solution:

Rewrite each term:

$$\frac{a}{b+c} = \frac{a^2}{ab+ac}$$

By Titu's Lemma:

$$\begin{aligned} \frac{a^2}{ab+ac} + \frac{b^2}{bc+ba} + \frac{c^2}{ca+cb} &\geq \frac{(a+b+c)^2}{ab+bc+ca} \\ &\geq \frac{(a+b+c)^2}{ab+bc+ca} \end{aligned}$$

Now prove:

$$\begin{aligned} \frac{(a+b+c)^2}{ab+bc+ca} &\geq \frac{3(a+b+c)^2}{2(ab+bc+ca)} \\ &\geq \frac{3(ab+bc+ca)}{2(ab+bc+ca)} \\ &\geq \frac{3}{2} \end{aligned}$$

This is true by the sum of squares inequality:

$$a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2] \geq 0$$

$$a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2] \geq 0$$

Example 3: Prove that for positive real numbers x, y, z :

$$xy+yz+zx+x+y+z \geq 3$$

This is Nesbitt's inequality, proven above using Cauchy-Schwarz.

Example 4: If $x_1, x_2, \dots, x_n$ are positive real numbers with sum S , prove:

$$\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \geq \frac{n^2}{S}$$

Solution:

By Cauchy-Schwarz (Titu's Lemma):

$$\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \geq \frac{(1+1+\dots+1)^2}{x_1 + x_2 + \dots + x_n} = \frac{n^2}{S}$$

Equality when all x_i are equal.

4.6 Applications of Cauchy-Schwarz

1. **Optimization:** Finding maximum/minimum values of linear expressions under quadratic constraints
2. **Vector Analysis:** Establishing properties of dot products and norms
3. **Probability Theory:** Proving that correlation coefficients are between -1 and 1
4. **Functional Analysis:** Foundation for Hilbert space theory
5. **Mathematical Competitions:** Essential for solving optimization problems

PAGE 5: JENSEN'S INEQUALITY

5. Jensen's Inequality

Jensen's inequality is a fundamental result about convex functions that generalizes many other classical inequalities. It provides a powerful method for proving inequalities using the geometric properties of convex functions.

5.1 Convex and Concave Functions

Definition: A function $f: I \rightarrow \mathbb{R}$ is **convex** on interval I if for all $x, y \in I$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

Geometric Interpretation: The line segment connecting any two points on the graph of f lies above or on the graph.

Definition: A function f is **concave** if $-f$ is convex, or equivalently:

$$f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y) \quad f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y)$$

Second Derivative Test: If f is twice differentiable:

- f is convex if $f''(x) \geq 0$ for all $x \in I$
- f is concave if $f''(x) \leq 0$ for all $x \in I$

Examples of Convex Functions:

- $f(x) = x^2$ (since $f''(x) = 2 > 0$)
- $f(x) = e^x$ (since $f''(x) = e^x > 0$)
- $f(x) = -\ln(x)$ for $x > 0$ (since $f''(x) = \frac{1}{x^2} > 0$)
- $f(x) = |x|$

Examples of Concave Functions:

- $f(x) = \ln(x)$ for $x > 0$ (since $f''(x) = -\frac{1}{x^2} < 0$)
- $f(x) = \sqrt{x}$ for $x \geq 0$ (since $f''(x) = -\frac{1}{4x^{3/2}} < 0$)
- $f(x) = -x^2$

5.2 Statement of Jensen's Inequality

Discrete Form: Let f be a convex function on interval I , and let $x_1, \dots, x_n \in I$ with weights $\lambda_1, \dots, \lambda_n \geq 0$ such that $\sum_{i=1}^n \lambda_i = 1$:

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i) \quad f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i)$$

Equal Weights Case:

$$f\left(\frac{x_1 + \dots + x_n}{n}\right) \leq \frac{f(x_1) + \dots + f(x_n)}{n} \quad f\left(\frac{x_1 + \dots + x_n}{n}\right) \leq \frac{f(x_1) + \dots + f(x_n)}{n}$$

If f is concave, the inequality reverses:

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \geq \sum_{i=1}^n \lambda_i f(x_i) \quad f\left(\sum_{i=1}^n \lambda_i x_i\right) \geq \sum_{i=1}^n \lambda_i f(x_i)$$

Equality Condition: Equality holds if and only if all x_i are equal, or f is linear on the interval containing all x_i .

5.3 Proof of Jensen's Inequality

Proof by Mathematical Induction:

Base Case: $n = 2$ is the definition of convexity.

Inductive Step: Assume true for $n = k$. For $n = k+1$:

Let $\lambda_1, \dots, \lambda_{k+1}$ be weights summing to 1. Let $\Lambda = \lambda_1 + \dots + \lambda_k = 1 - \lambda_{k+1}$.

If $\Lambda = 0$, the result is trivial. Otherwise:

$$f(\sum_{i=1}^{k+1} \lambda_i x_i) = f(\Lambda \sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1}) = f(\Lambda \sum_{i=1}^k \lambda_i x_i + \lambda_{k+1} x_{k+1})$$

By convexity (base case $n=2$):

$$\leq \Lambda f(\sum_{i=1}^k \lambda_i x_i) + \lambda_{k+1} f(x_{k+1}) \leq \Lambda f(\sum_{i=1}^k \lambda_i x_i) + \lambda_{k+1} f(x_{k+1})$$

By induction hypothesis:

$$\leq \Lambda \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i) \leq \Lambda \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i)$$

5.4 Deriving AM-GM from Jensen's Inequality

Since $f(x) = -\ln(x)$ is convex for $x > 0$:

$$-\ln\left(\frac{a_1 + \dots + a_n}{n}\right) \leq -\ln\left(\frac{a_1}{n}\right) + \dots + (-\ln\left(\frac{a_n}{n}\right)) = -\ln\left(\frac{a_1 \dots a_n}{n^n}\right) - \ln(n)$$

$$+ \dots + \ln(n) \leq -\ln(a_1) + \dots + (-\ln(a_n)) = -\ln(a_1 \dots a_n)$$

Taking negative:

$$\ln\left(\frac{a_1 + \dots + a_n}{n}\right) \geq \ln\left(\frac{a_1 \dots a_n}{n^n}\right) \ln(n) \geq \ln(a_1 \dots a_n)$$

Exponentiating (since e^x is increasing):

$$\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}$$

5.5 Worked Examples

Example 1: Prove that for positive real numbers a, b, c :

$$\ln(a+b+c) \geq \ln(a) + \ln(b) + \ln(c) \quad 3\ln(3a+b+c) \geq 3\ln(a) + \ln(b) + \ln(c)$$

Solution:

Since $f(x) = \ln(x)$ is concave (not convex), by Jensen's inequality for concave functions:

$$\ln\left(\frac{a+b+c}{3}\right) \geq \frac{\ln(a) + \ln(b) + \ln(c)}{3} \quad 3\ln\left(\frac{a+b+c}{3}\right) \geq \ln(a) + \ln(b) + \ln(c)$$

Taking exponentials:

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc} \quad 3a+b+c \geq abc$$

This is AM-GM.

Example 2: If $x_1, \dots, x_n > 0$, prove:

$$e^{x_1} + \dots + e^{x_n} \geq e^{x_1 + \dots + x_n} \quad e^{x_1} + \dots + e^{x_n} \geq e^{x_1 + \dots + x_n}$$

Solution:

Since $f(x) = e^x$ is convex ($f'(x) = e^x > 0$), by Jensen's inequality:

$$\frac{e^{x_1} + \dots + e^{x_n}}{n} \geq e^{\frac{x_1 + \dots + x_n}{n}} \Rightarrow e^{x_1} + \dots + e^{x_n} \geq n e^{\frac{x_1 + \dots + x_n}{n}} = e^{x_1 + \dots + x_n}$$

Example 3: For a triangle with angles A, B, C , prove:

$$\sin(A) + \sin(B) + \sin(C) \leq \frac{3\sqrt{3}}{2}$$

Solution:

Since $f(x) = \sin(x)$ is concave on $[0, \pi]$ ($f'(x) = \cos(x) \leq 0$ for $x \in [\frac{\pi}{2}, \pi]$):

$$\sin(A) + \sin(B) + \sin(C) \leq \sin\left(\frac{A+B+C}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

Therefore:

$$\sin(A) + \sin(B) + \sin(C) \leq \frac{3\sqrt{3}}{2}$$

Equality when $A = B = C = \frac{\pi}{3}$ (equilateral triangle).

Example 4: Prove that for positive real numbers a, b, c :

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

Solution using Jensen:

Let $f(x) = \frac{1}{x}$, which is convex for $x > 0$ ($f'(x) = -\frac{1}{x^2} < 0$).

Let $s = a + b + c$. Then:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$$

By Jensen's inequality applied to $f(x) = \frac{1}{x}$ (which is convex):

$$\frac{1}{3} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq \frac{1}{\frac{a+b+c}{3}} \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{a+b+c}$$

Therefore:

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

5.6 Applications of Jensen's Inequality

1. **Proving AM-GM:** As shown above
2. **Information Theory:** Proving properties of entropy and KL divergence

3. **Probability Theory:** Proving that $E[f(X)] \geq f(E[X])$ for convex f
4. **Economics:** Proving properties of utility functions
5. **Mathematical Competitions:** Powerful tool for proving complex inequalities

PAGE 6: POWER MEAN INEQUALITY

6. Power Mean Inequality (QM-AM-GM-HM)

The Power Mean Inequality establishes a hierarchy among different types of means, showing that as the power increases, the mean also increases.

6.1 Definition of Means

For k positive real numbers $a_1, a_2, \dots, a_k$:

Arithmetic Mean (AM):

$$f_{AM} = \frac{a_1 + a_2 + \dots + a_k}{k} \quad f_{AM} = \frac{1}{k}(a_1 + a_2 + \dots + a_k)$$

Geometric Mean (GM):

$$f_{GM} = \sqrt[k]{a_1 a_2 \dots a_k} \quad f_{GM} = \sqrt[k]{a_1 a_2 \dots a_k}$$

Harmonic Mean (HM):

$$f_{HM} = \frac{k}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}} \quad f_{HM} = \frac{1}{\frac{1}{k}(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k})}$$

Quadratic Mean (QM) or Root Mean Square:

$$f_{QM} = \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_k^2}{k}} \quad f_{QM} = \sqrt{\frac{1}{k}(a_1^2 + a_2^2 + \dots + a_k^2)}$$

6.2 General Power Mean

For positive numbers $a_1, \dots, a_n$ and real number r :

$$M_r(a_1, \dots, a_n) = \begin{cases} \left(\frac{a_1^r + \dots + a_n^r}{n} \right)^{1/r}, & r \neq 0 \\ \sqrt[n]{a_1 \dots a_n}, & r = 0 \end{cases}$$

Special Cases:

- M_{-1} = Harmonic Mean
- M_0 = Geometric Mean (limit as $r \rightarrow 0$)
- M_1 = Arithmetic Mean
- M_2 = Quadratic Mean

6.3 Statement of the Power Mean Inequality

For positive real numbers $a_1, \dots, a_n$ and real numbers $r < s$:

$$M_r(a_1, \dots, a_n) \leq M_s(a_1, \dots, a_n)$$

with equality if and only if $a_1 = a_2 = \dots = a_n$.

6.4 The Famous Chain: $QM \geq AM \geq GM \geq HM$

For positive real numbers $a_1, \dots, a_n$:

$$\frac{a_1^2 + \dots + a_n^2}{n} \geq \frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n} \geq \frac{1}{\frac{1}{a_1} + \dots + \frac{1}{a_n}}$$

6.5 Proofs of Individual Inequalities

QM $\geq$ AM

Proof:

By Cauchy-Schwarz:

$$(1^2 + 1^2 + \dots + 1^2)(a_1^2 + a_2^2 + \dots + a_n^2) \geq (a_1 + a_2 + \dots + a_n)^2$$

$$n(a_1^2 + \dots + a_n^2) \geq (a_1 + \dots + a_n)^2$$

$$\frac{a_1^2 + \dots + a_n^2}{n} \geq \left(\frac{a_1 + \dots + a_n}{n}\right)^2$$

Taking square roots:

$$\frac{a_1^2 + \dots + a_n^2}{n} \geq \left(\frac{a_1 + \dots + a_n}{n}\right)^2 \Rightarrow \frac{a_1^2 + \dots + a_n^2}{n} \geq \left(\frac{a_1 + \dots + a_n}{n}\right)^2$$

AM $\geq$ GM

Proven in Section 3 using various methods.

GM $\geq$ HM

Proof:

Apply AM-GM to the reciprocals $\frac{1}{a_1}, \frac{1}{a_2}, \dots, \frac{1}{a_n}$:

$$\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \geq \frac{1}{a_1} \cdot \frac{1}{a_2} \cdot \dots \cdot \frac{1}{a_n} = \frac{1}{a_1 a_2 \dots a_n}$$

Taking reciprocals (which reverses the inequality):

$$\frac{1}{\frac{1}{a_1} + \dots + \frac{1}{a_n}} \leq \frac{1}{\frac{1}{a_1} \cdot \frac{1}{a_2} \cdot \dots \cdot \frac{1}{a_n}} = \frac{a_1 a_2 \dots a_n}{n}$$

Therefore, $GM \geq HM$.

6.6 Proof Using Jensen's Inequality

The Power Mean Inequality follows from the convexity of $f(x) = x^{\{s/r\}}$ for $s > r > 0$:

Let $r < s$. Consider $f(x) = x^{\{s/r\}}$, which is convex for $x > 0$ since $s/r > 1$.

By Jensen's inequality:

$$\frac{(a_1^r + \dots + a_n^r)^{s/r}}{n} \leq \frac{(a_1^s + \dots + a_n^s)^{1/s}}{n} \quad \text{or} \quad (a_1^r + \dots + a_n^r)^{1/r} \leq (a_1^s + \dots + a_n^s)^{1/s}$$

Taking the s -th root:

$$(a_1^r + \dots + a_n^r)^{1/r} \leq (a_1^s + \dots + a_n^s)^{1/s} \quad \text{or} \quad (a_1^r + \dots + a_n^r)^{1/r} \leq (a_1^s + \dots + a_n^s)^{1/s}$$

Therefore, $M_r \leq M_s$.

6.7 Worked Examples

Example 1: Prove that for positive real numbers a, b, c :

$$a^2 + b^2 + c^2 \geq a + b + c \geq abc \geq \frac{1}{3}a + \frac{1}{3}b + \frac{1}{3}c \geq \frac{1}{3}a + \frac{1}{3}b + \frac{1}{3}c \geq \frac{1}{3}a + \frac{1}{3}b + \frac{1}{3}c$$

Solution:

This is the QM-AM-GM-HM inequality for $n = 3$, which follows directly from the Power Mean Inequality.

Example 2: If $x_1, \dots, x_n > 0$, prove:

$$x_1^2 + \dots + x_n^2 \geq \frac{(x_1 + \dots + x_n)^2}{n} \geq \frac{(x_1 + \dots + x_n)^2}{n}$$

Solution:

This is $QM \geq AM$ squared:

$$x_1^2 + \dots + x_n^2 \geq \frac{(x_1 + \dots + x_n)^2}{n} \geq \frac{(x_1 + \dots + x_n)^2}{n}$$

Squaring both sides:

$$x_1^2 + \dots + x_n^2 \geq \frac{(x_1 + \dots + x_n)^2}{n} \geq \frac{(x_1 + \dots + x_n)^2}{n}$$

Example 3: Find the minimum value of $x^2 + y^2 + z^2$ given $x + y + z = 1$ and $x, y, z > 0$.

Solution:

By $QM \geq AM$:

$$x^2 + y^2 + z^2 \geq \frac{(x + y + z)^2}{3} = \frac{1^2}{3} = \frac{1}{3}$$

Squaring:

$$x^2 + y^2 + z^2 \geq \frac{1}{3}(x^2 + y^2 + z^2) \geq \frac{1}{3} \cdot 9 = 3$$

$$x^2 + y^2 + z^2 \geq 3 \cdot \frac{1}{3} = 1$$

Minimum value is $\frac{1}{3}$, achieved when $x = y = z = \frac{1}{3}$.

Example 4: Prove that for positive real numbers a, b, c :

$$a^3 + b^3 + c^3 \geq 3abc$$

Solution:

This is AM $\geq$ HM, which follows from the Power Mean Inequality ($M_1 \geq M_{-1}$).

6.8 Applications of Power Mean Inequality

1. **Optimization:** Finding bounds on sums of powers
2. **Statistics:** Understanding relationships between different measures of central tendency
3. **Physics:** Establishing relationships between different types of averages
4. **Economics:** Comparing different welfare measures
5. **Mathematical Competitions:** Fundamental tool for inequality problems

PAGE 7: HÖLDER'S AND MINKOWSKI'S INEQUALITIES

7. Hölder's and Minkowski's Inequalities

These inequalities generalize the Cauchy-Schwarz inequality and are fundamental in functional analysis and measure theory.

7.1 Hölder's Inequality

Statement

For non-negative sequences $x_1, \dots, x_n$ and $y_1, \dots, y_n$, with $p, q > 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$ (conjugate exponents):

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n x_i^p \right)^{1/p} \left(\sum_{i=1}^n y_i^q \right)^{1/q}$$

Equality Condition: Equality holds if and only if the sequences are proportional, i.e., there exist constants $\alpha, \beta \geq 0$ (not both zero) such that $\alpha x_i^p = \beta y_i^q$ for all i .

Special Case: When $p = q = 2$, Hölder's inequality reduces to the Cauchy-Schwarz inequality.

Proof Using Weighted AM-GM

By the weighted AM-GM inequality with weights $\frac{1}{p}$ and $\frac{1}{q}$:

$$x_i p \sum x_j p \cdot 1 p + y_i q \sum y_j q \cdot 1 q \geq (x_i p \sum x_j p)^{1/p} (y_i q \sum y_j q)^{1/q} = x_i y_i (\sum x_j p)^{1/p} (\sum y_j q)^{1/q} \sum x_j p x_i p \cdot p 1 + \sum y_j q y_i q \cdot q 1 \geq (\sum x_j p x_i p)^{1/p} (\sum y_j q y_i q)^{1/q} = (\sum x_j p)^{1/p} (\sum y_j q)^{1/q} x_i y_i$$

Summing over i and using $\frac{1}{p} + \frac{1}{q} = 1$:

$$1 p + 1 q = 1 \geq \sum x_i y_i (\sum x_j p)^{1/p} (\sum y_j q)^{1/q} p 1 + q 1 = 1 \geq (\sum x_j p)^{1/p} (\sum y_j q)^{1/q} \sum x_i y_i$$

Therefore:

$$\sum_{i=1}^n x_i y_i \leq (\sum_{i=1}^n x_i p)^{1/p} (\sum_{i=1}^n y_i q)^{1/q} \sum_{i=1}^n x_i y_i \leq (\sum_{i=1}^n x_i p)^{1/p} (\sum_{i=1}^n y_i q)^{1/q}$$

General Form for Multiple Sequences

For k sequences and exponents $p_1, \dots, p_k > 1$ with $\sum_{j=1}^k \frac{1}{p_j} = 1$:

$$\sum_{i=1}^n \prod_{j=1}^k x_{ij} \leq \prod_{j=1}^k (\sum_{i=1}^n x_{ij}^{p_j})^{1/p_j} \sum_{i=1}^n \prod_{j=1}^k x_{ij} \leq \prod_{j=1}^k (\sum_{i=1}^n x_{ij}^{p_j})^{1/p_j}$$

7.2 Minkowski's Inequality

Statement

For non-negative numbers $x_1, \dots, x_n$ and $y_1, \dots, y_n$ with $p \geq 1$:

$$(\sum_{i=1}^n (x_i + y_i)^p)^{1/p} \leq (\sum_{i=1}^n x_i^p)^{1/p} + (\sum_{i=1}^n y_i^p)^{1/p} (\sum_{i=1}^n (x_i + y_i)^p)^{1/p} \leq (\sum_{i=1}^n x_i^p)^{1/p} + (\sum_{i=1}^n y_i^p)^{1/p}$$

This is the triangle inequality for the p -norm.

Equality Condition: Equality holds if and only if the sequences are proportional (one is a non-negative scalar multiple of the other).

Proof Using Hölder's Inequality

For $p = 1$, the inequality is trivial. For $p > 1$:

$$\sum_{i=1}^n (x_i + y_i)^p = \sum_{i=1}^n x_i (x_i + y_i)^{p-1} + \sum_{i=1}^n y_i (x_i + y_i)^{p-1} \sum_{i=1}^n (x_i + y_i)^p = \sum_{i=1}^n x_i (x_i + y_i)^{p-1} + \sum_{i=1}^n y_i (x_i + y_i)^{p-1}$$

Applying Hölder's inequality to each sum with exponents p and $q = \frac{p}{p-1}$:

$$\sum_{i=1}^n x_i (x_i + y_i)^{p-1} \leq (\sum_{i=1}^n x_i^p)^{1/p} (\sum_{i=1}^n (x_i + y_i)^p)^{1/q} \sum_{i=1}^n x_i (x_i + y_i)^{p-1} \leq (\sum_{i=1}^n x_i^p)^{1/p} (\sum_{i=1}^n (x_i + y_i)^p)^{1/q}$$

$$\sum_{i=1}^n y_i (x_i + y_i)^{p-1} \leq (\sum_{i=1}^n y_i^p)^{1/p} (\sum_{i=1}^n (x_i + y_i)^p)^{1/q} \sum_{i=1}^n y_i (x_i + y_i)^{p-1} \leq (\sum_{i=1}^n y_i^p)^{1/p} (\sum_{i=1}^n (x_i + y_i)^p)^{1/q}$$

Adding:

$$\sum_{i=1}^n (x_i + y_i)^p \leq [(\sum_{i=1}^n x_i^p)^{1/p} + (\sum_{i=1}^n y_i^p)^{1/p}] (\sum_{i=1}^n (x_i + y_i)^p)^{1/q} \sum_{i=1}^n (x_i + y_i)^p \leq [(\sum_{i=1}^n x_i^p)^{1/p} + (\sum_{i=1}^n y_i^p)^{1/p}] (\sum_{i=1}^n (x_i + y_i)^p)^{1/q}$$

Dividing by $(\sum_{i=1}^n (x_i + y_i)^p)^{1/q}$ and noting $1 - \frac{1}{q} = \frac{1}{p}$:

$$(\sum_{i=1}^n (x_i + y_i)^p)^{1/p} \leq (\sum_{i=1}^n x_i^p)^{1/p} + (\sum_{i=1}^n y_i^p)^{1/p} (\sum_{i=1}^n (x_i + y_i)^p)^{1/p} \leq (\sum_{i=1}^n x_i^p)^{1/p} + (\sum_{i=1}^n y_i^p)^{1/p}$$

7.3 Young's Inequality

Statement

For non-negative real numbers a, b and positive real numbers p, q satisfying $\frac{1}{p} + \frac{1}{q} = 1$:

$$ab \leq ap + bq \quad a^p + b^q \leq pa + qb$$

Equality Condition: Equality holds if and only if $a^p = b^q$.

Proof

By weighted AM-GM:

$$ab = (ap)^{1/p} (bq)^{1/q} \leq ap + bq \quad a^p + b^q = (ap)^{1/p} (bq)^{1/q} \leq pa + qb$$

This is a special case of the weighted AM-GM inequality.

7.4 Worked Examples

Example 1: Prove that for positive real numbers a, b, c :

$$a^3 + b^3 + c^3 \geq a + b + c \quad a^3 + b^3 + c^3 \geq 3a + 3b + 3c$$

Solution:

By Minkowski's inequality with $p = 3$:

$$((a^3)^{1/3} + (b^3)^{1/3} + (c^3)^{1/3})^3 \geq a + b + c \quad 2^3/3 = a + b + c \quad 9(a^3 + b^3 + c^3)^{1/3} \geq 3a + 3b + 3c$$

This is a direct application of Minkowski's inequality.

Example 2: Show that Hölder's inequality generalizes Cauchy-Schwarz.

Solution:

When $p = q = 2$, we have $\frac{1}{2} + \frac{1}{2} = 1$.

Hölder's inequality becomes:

$$\sum_{i=1}^n x_i y_i \leq (\sum_{i=1}^n x_i^2)^{1/2} (\sum_{i=1}^n y_i^2)^{1/2}$$

Squaring both sides:

$$(\sum_{i=1}^n x_i y_i)^2 \leq (\sum_{i=1}^n x_i^2) (\sum_{i=1}^n y_i^2)$$

This is exactly the Cauchy-Schwarz inequality.

Example 3: Prove that for $p \geq 1$:

$$\|x+y\|_p \leq \|x\|_p + \|y\|_p$$

where $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$.

Solution:

This is exactly Minkowski's inequality, which states that the p -norm satisfies the triangle inequality.

7.5 Applications

1. **Functional Analysis:** Foundation for L^p spaces and Banach spaces
2. **Measure Theory:** Fundamental in integration theory
3. **Probability Theory:** Establishing bounds on moments
4. **Partial Differential Equations:** Estimating solutions
5. **Signal Processing:** Analyzing signals in different norms

PAGE 8: REARRANGEMENT AND CHEBYSHEV'S INEQUALITIES**8. Rearrangement and Chebyshev's Inequalities**

These inequalities deal with the ordering of sequences and are closely related to each other.

8.1 Rearrangement Inequality**Statement**

Given two sequences $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$, for any permutation σ of $\{1, 2, \dots, n\}$:

$$a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1 \leq a_1 b_{\sigma(1)} + a_2 b_{\sigma(2)} + \dots + a_n b_{\sigma(n)} \leq a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

$$+ a_2 b_{n-1} + \dots + a_n b_1 \leq a_1 b_{\sigma(1)} + a_2 b_{\sigma(2)} + \dots + a_n b_{\sigma(n)} \leq a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

In other words:

- The sum is **maximized** when sequences are similarly ordered (both increasing or both decreasing)
- The sum is **minimized** when sequences are oppositely ordered

Equality Condition: Equality in the upper bound holds if and only if the sequences are similarly ordered. Equality in the lower bound holds if and only if the sequences are oppositely ordered.

Proof for $n = 2$

For $a_1 \leq a_2$ and $b_1 \leq b_2$:

$$a_1 b_1 + a_2 b_2 - (a_1 b_2 + a_2 b_1) = a_1(b_1 - b_2) + a_2(b_2 - b_1) = (a_2 - a_1)(b_2 - b_1) \geq 0$$

$$a_1 b_1 + a_2 b_2 \geq a_1 b_2 + a_2 b_1$$

Therefore:

$$a_1 b_1 + a_2 b_2 \geq a_1 b_2 + a_2 b_1$$

The general case uses the same idea by swapping adjacent elements.

8.2 Chebyshev's Inequality

Statement

Given non-increasing sequences $\{a_i\}$ and $\{b_i\}$:

$$\sum_{i=1}^n a_i b_i \geq \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \geq \frac{1}{n} \sum_{i=1}^n a_i b_{n+1-i}$$

Equivalent Form:

$$\sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \geq \sum_{i=1}^n a_i b_{n+1-i}$$

Equality Condition: Equality holds if and only if all a_i are equal or all b_i are equal.

Proof Using Rearrangement Inequality

By the rearrangement inequality:

$$\sum_{i=1}^n a_i b_i \geq \sum_{i=1}^n a_i b_{\sigma(i)} \geq \sum_{i=1}^n a_i b_{\sigma(i)}$$

for any permutation σ . Summing over all $n!$ permutations:

$$n! \sum_{i=1}^n a_i b_i \geq \sum_{\sigma} \sum_{i=1}^n a_i b_{\sigma(i)} = (n-1)! \sum_{i=1}^n a_i \sum_{j=1}^n b_j n! \sum_{i=1}^n a_i b_i \geq \sum_{\sigma} \sum_{i=1}^n a_i b_{\sigma(i)} \\ = (n-1)! \sum_{i=1}^n a_i \sum_{j=1}^n b_j$$

Dividing by $n!$:

$$\sum_{i=1}^n a_i b_i \geq \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right)$$

Dividing by n :

$$\sum_{i=1}^n a_i b_i \geq \frac{1}{n} \sum_{i=1}^n a_i \times \frac{1}{n} \sum_{i=1}^n b_i \geq \frac{1}{n} \sum_{i=1}^n a_i \times \frac{1}{n} \sum_{i=1}^n b_i$$

8.3 Worked Examples

Example 1: Prove Nesbitt's Inequality using Chebyshev's inequality:

For positive real numbers a, b, c :

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

Solution:

Without loss of generality, assume $a \geq b \geq c$. Then $\frac{1}{b+c} \leq \frac{1}{a+c} \leq \frac{1}{a+b}$.

By Chebyshev's inequality:

$$3(ab+bc+ca) \geq (a+b+c)(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b}) \geq (a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right) \geq (a+b+c) \left(\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \right)$$

By Titu's lemma:

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{(a+b+c)^2}{a(b+c) + b(c+a) + c(a+b)} \geq \frac{(a+b+c)^2}{2(ab+bc+ca)} \geq \frac{3}{2}$$

Therefore:

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$

Example 2: If $a_1, a_2, \dots, a_n$ are distinct positive real numbers, prove:

$$\frac{a_1}{a_2+a_3+\dots+a_n} + \frac{a_2}{a_1+a_3+\dots+a_n} + \dots + \frac{a_n}{a_1+a_2+\dots+a_{n-1}} \geq \frac{n}{n-1}$$

Solution:

By AM-GM:

$$\frac{a_1}{a_2+a_3+\dots+a_n} + \frac{a_2}{a_1+a_3+\dots+a_n} + \dots + \frac{a_n}{a_1+a_2+\dots+a_{n-1}} \geq \frac{n}{n-1} \left(\frac{a_1}{a_2+a_3+\dots+a_n} + \frac{a_2}{a_1+a_3+\dots+a_n} + \dots + \frac{a_n}{a_1+a_2+\dots+a_{n-1}} \right) \geq \frac{n}{n-1}$$

Alternatively, by the rearrangement inequality, the sum is minimized when the sequences are similarly ordered.

Example 3: Prove that for any permutation σ of $\{1, 2, \dots, n\}$:

$$\sum_{i=1}^n i \sigma(i) \geq \sum_{i=1}^n i^2$$

Solution:

The sequences $\{1, 2, \dots, n\}$ and $\left\{\frac{1}{1}, \frac{1}{2}, \dots, \frac{1}{n}\right\}$ are oppositely ordered. By the rearrangement inequality, the sum is minimized when they are similarly ordered, giving:

$$\sum_{i=1}^n i \cdot i = \sum_{i=1}^n i \cdot 1 = n \sum_{i=1}^n i = \sum_{i=1}^n i \cdot 1 = n$$

For any other permutation, the sum is at least n .

8.4 Applications

1. **Combinatorial Optimization:** Finding optimal permutations
2. **Mathematical Competitions:** Essential for olympiad problems
3. **Statistics:** Understanding correlations between ordered data
4. **Economics:** Analyzing wealth distribution and inequality

PAGE 9: ADVANCED APPLICATIONS AND FAMOUS PROBLEMS**9. Advanced Applications and Famous Problems**

Classical inequalities are essential tools for solving complex mathematical problems, particularly in competitions and research.

9.1 Nesbitt's Inequality**Statement**

For positive real numbers a, b, c :

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$$